Evaluation of Speaker Anonymization on Emotional Speech

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A. EXTENDED SUMMARY

a) Introduction

Since speech data carries a range of personal information, such as gender and health information, which may make a speaker identifiable, speech privacy-preserving solutions are becoming increasingly important. Additionally, the General Data Protection Regulation (GDPR) emphasizes the importance of protecting personal data.

In this article, the authors evaluate the Voice Privacy Challenge (VPC) baseline system, which represents one of the first attempts by the speech community to anonymize speakers. The evaluation uses the Interactive Emotional Dyadic Motion Capture (IEMOCAP) dataset, which contains both scripted and improvised interactions. Although the dataset includes only a small number of speakers, it provides varied emotional expressions under semi-realistic conditions.

There are multiple ways to analyse linguistic cues. Speech emotion recognition (SER) considers prosodic features such as fundamental frequency (F0), intensity and rhythm. Another option is to extract Mel-frequency cepstral coefficients (MFCCs) or spectrograms. The HUMAINE Association also compiled an inventory of interpretable acoustic features through the CEICES initiative. However, extracting these features from degraded signals can be error-prone. Nowadays, pre-trained features are widely used, particularly embeddings extracted by neural models trained for speech-processing tasks other than SER.

Previous studies have used techniques such as homomorphic encryption, hashing, and generative adversarial networks to protect speech privacy. In this paper, the authors investigate how applying an anonymization process on emotional data, which is supposed to hide the speaker identity, impacts SER performance.

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b) Method

b.1) Anonymization framework

The aim of the VPC baseline is to convert the original voice into another voice to hide the identity of the original speaker. The anonymization system depicted in Figure 1 can be decomposed into three groups of modules.

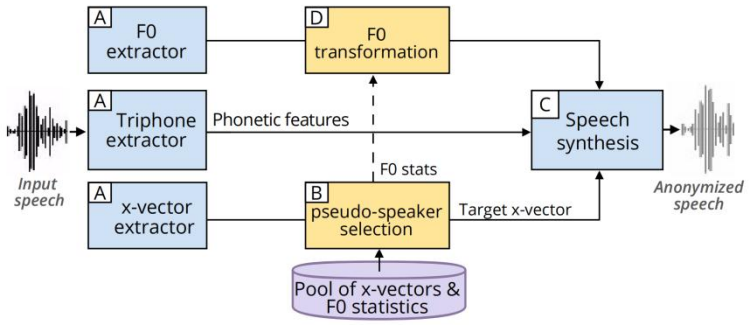


Figure 1: The Voice Privacy speaker anonymization pipeline. Modules A, B, and C are parts of the baseline model. Module D is an enhancement used to transform the F0 values

Modules in Group A extract different features from the source signal: the fundamental frequency (F0), the phonetic features encoding articulation of speech sounds and the speaker's x-vector; Module B derives a new pseudo-speaker target x-vector identity. It compares the source speaker's x-vector with a pool of external x-vectors and selects the 200 furthest x-vectors. From these, 100 x-vectors are randomly selected and averaged to create the anonymized pseudo-speaker x-vector. Finally, Module C synthesizes a speech waveform using the target x-vector together with the original phonetic features and F0.

Previous studies have found that modifying F0 can improve the quality of converted speech. Therefore, the authors propose to modify the F0 values of a source utterance from a given speaker (cf. module D in Figure 1) by using a linear transformation and a random warping.

(1) **Linear transformation**: Aims to adjust the source F0 values so that they are closer to those of the selected pseudo-speaker. It is defined as:

$$\log \hat{x}_t = \mu_y + \frac{\sigma_y}{\sigma_x}(\log x_t - \mu_x)$$

where x_t represents the F0 of the source speaker at frame t, and $\hat{x}_t$ represents the transformed F0. The parameters μ_x and σ_x are the mean and standard deviation of the source speaker's log-scaled F0, while μ_y and σ_y are the corresponding mean and standard deviation of the target pseudo-speaker. These statistics are calculated using voiced frames only. The target statistics are obtained by averaging the F0 statistics of the 100 speakers selected to construct the pseudo-speaker x-vector.

(2) **Random Warping:** After the linear transformation, random warping is applied to increase or decrease the range of F0 variation. It is defined as:

$$\hat{x}_t = \mu_x + (x_t - \mu_x) \times \alpha$$

where x_t is the F0 value at frame t, μ_x is the mean F0 of the utterance, and α is a warping factor randomly sampled from a uniform distribution between 0.8 and 1.2 for each utterance. A value of α below 1 reduces the F0 variation, whereas a value above 1 increases it.

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b.2) The VPC attack scenarios

In this work, the authors focus on the Ignorant and Informed attacker scenarios. The main difference between the two scenarios is that, in the former, privacy is assessed using models trained on original, non-anonymized data, whereas the latter uses models trained on data anonymized in the same manner as the service provider's data. The evaluation data are anonymized in both scenarios.

b.3) Evaluation experiment

The IEMOCAP dataset is used in the experiment. It consists of 12 hours of improvised and scripted English dialogues performed by ten actors, five male and five female. Directional microphones are used to capture each speaker's speech. When two speakers overlap, the speaker closest to the microphone is considered dominant.

The data are segmented by dominant speaker turns. Each turn is annotated with emotion categories by six human annotators. Only recordings for which the majority of annotators agreed on the emotion are used. The authors consider five emotions: neutral, frustration, sadness, anger, and happiness. The happiness category combines the original happiness and excitement annotations to balance the number of utterances in each emotion class.

In this paper, the authors evaluate the emotional information present in both the original utterances and their corresponding anonymized versions using a Support Vector Machine (SVM). The SVM uses a radial basis function kernel and eGeMAPS acoustic features, which provide a compact representation of emotional information. The authors also test MFCC features and obtain similar results. Following the VPC attack scenarios, the SVM is trained on original speech in the Ignorant scenario and on anonymized speech in the Informed scenario.

Emotion-recognition performance is measured using Unweighted Average Recall (UAR), where a higher value indicates better recognition performance. Because the dataset is relatively small, leave-one-session-out cross-validation is used, and the final UAR is calculated across the five test folds. The utility of the anonymized speech is

evaluated using Automatic Speech Recognition (ASR). Performance is measured using the Word Error Rate (WER), where a lower value indicates that the linguistic content is more intelligible. Two ASR systems are used: one trained on original speech and another trained on anonymized speech.

c) Research Results

The results show that the Ignorant attacker achieves a UAR of only 21.97%, which is close to random guessing. However, when the attacker is trained on similarly anonymized speech, the UAR increases to 37.92%, representing only a 15% relative reduction compared with the 44.48% obtained on original speech. Furthermore, the linear F0 transformation and random warping produce almost no additional reduction in emotion-recognition performance. These findings indicate that the VPC baseline conceals speaker identity but does not effectively remove emotional information from speech.

d) Feedback

I believe that evaluating both Ignorant and Informed attacker scenarios is useful for demonstrating the risk of emotional information leakage under different attacker capabilities. However, these two scenarios do not completely isolate the effect of the attacker's knowledge. The Ignorant attacker is trained on original speech but tested on anonymized speech. Therefore, its lower recognition performance may partly result from the distribution mismatch between the training and test data. In contrast, the Informed attacker is trained and tested on anonymized speech, allowing the model to adapt to the feature changes caused by anonymization. Consequently, the performance difference between the two scenarios cannot be attributed solely to whether the attacker understands the anonymization method. Future research could conduct more controlled experiments using data from the same source and of the same size while varying only the anonymization information available to the attacker. This would help distinguish the respective effects of attacker knowledge, data distribution, and model adaptation.

This study also has several other limitations. First, the experiment uses only the IEMOCAP dataset, which contains English speech from only ten actors. Therefore, it may not fully represent different languages, accents, speakers, and forms of emotional expression in real-world environments. Second, the paper evaluates emotional information using only an SVM with eGeMAPS features. A stronger neural emotion-recognition model might recover more emotional information from anonymized speech. Therefore, the current results may underestimate the risk of emotional information leakage.

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B. DESIGN / IMPROVEMENT OF SECURITY SYSTEM

The question I select is *(1) Design an improved security system based on the existing systems in the paper.*

Solution 1: emotion-hiding mode
Since the main goal of the VPC baseline is to conceal the speaker identity while preserving the linguistic content, and current experiments have shown that emotional information has not been well concealed, we can make the following adjustments based on the structure of the VPC baseline.
1. Normalize multiple prosodic features: The original F0 transformation could be extended to jointly normalize multiple prosodic features, including pitch contour, intensity, rhythm, speaking rate, and duration. This is necessary because emotional information is not carried by F0 alone.
2. Emotion-neutral paraphrasing: The system could use automatic speech recognition to obtain the transcript and apply emotion-neutral paraphrasing before speech synthesis. Emotional words and degree adverbs could be removed or weakened while the core meaning of the utterance is preserved.

Solution 2: emotion-preserving privacy mode
This approach complements Solution 1. Rather than suppressing emotional information, it **preserves the emotional cues** required by the service while preventing the original speech signal from being transmitted to the server. This is important because identical sentences may express different meanings or attitudes depending on the speaker's prosody.

In the proposed system, automatic speech recognition (ASR) and speech emotion recognition (SER) would be performed locally on the **user's device**. The ASR module would convert the speech into a standardized sentence and remove personally identifiable information, while the SER module would generate a coarse emotion label, an emotion-intensity score, and a confidence score. Only this privacy-minimized representation would be encrypted and transmitted to the server. The original speech signal and the speaker's x-vector would remain on the local device and be deleted after processing.

After receiving the standardized sentence and emotional information, the server could perform the required analysis or use a fixed **pseudo-speaker text-to-speech system** to reconstruct the message. This approach would reduce the disclosure of biometric voice information while preserving the emotional context required by the service. However, because emotional labels are also sensitive personal information, encrypted transmission, access control, and data-retention limits would still be necessary. The system should be evaluated in terms of speech-emotion accuracy, semantic preservation, identity leakage, communication cost, and processing latency.

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